Kinetic Theory of Gases

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Pressure of Gas

INTRODUCTION

A gas consists of atoms both individually or bound together as molecules that fill their container's volume and exert pressure on the container's walls. We can usually assign a temperature to such a contained gas. These three variables associated with gas temperature, volume, and pressure are all due to the motion of the atoms. The volume is a result of the freedom that atoms have to spread throughout the container, and the pressure is a result of the collisions of the atoms with the container's walls, and the temperature has to do with the kinetic energy of the atoms. The focus of this chapter will be to find the relation between measurable properties of gases such as pressure, temperature with molecular size, masses, and velocity distribution.
This chapter introduces the kinetic theory, which was developed in the nineteenth century by Maxwell Boltzmann and others.

KINETIC THEORY OF GASES:

The kinetic theory of gases is based on the following basic assumptions.

(a) Molecules undergo elastic collision with other molecules and the walls but otherwise exert no forces on each other. Molecules are so small that the volume of molecules is negligible as compared with the volume of the gas. Also, the number of gas molecules is very large.

(b) The direction of the motion of the gas Molecules is completely random.

(c) The molecules travel in straight lines.

(d) The time of the collision between any two molecules is very short.

(e) The collision between the wall of the container and molecules is perfectly elastic. It means kinetic energy is conserved in each collision.

(f) The path traveled by a molecule between two collisions is called free path, and the mean of this distance traveled by a molecule is called the mean free path.

(g) Newton's law of motion is applicable in the motion of molecules.

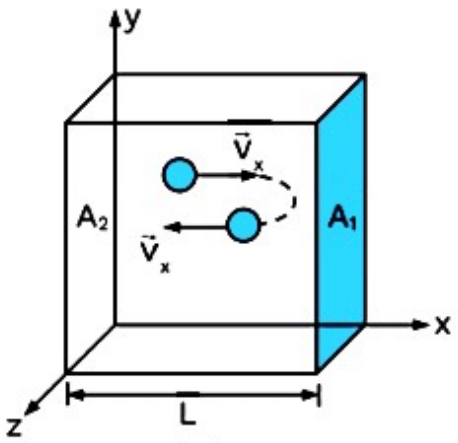
(h) There is no effect of gravity on the motion of molecules.

EXPRESSION FOR PRESSURE OF AN IDEAL GAS:

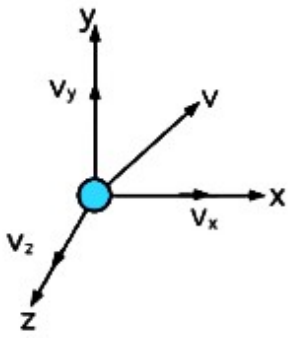
Consider a collection of N molecules of an ideal gas in a cubical container with edges of length L.
Let us consider that one of these molecules of mass m is moving with velocity $\vec{v}$ in any direction.
Then $\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$, where $v = \sqrt{v_x^2 + v_y^2 + v_z^2}$.

Due to random motion of molecule, $v_x = v_y = v_z$

$\Rightarrow v^2 = 3v_x^2 = 3v_y^2 = 3v_z^2$



(a)



(b)

When the molecule collides elastically with any wall, its velocity component perpendicular to the wall is reversed.

The change in momentum of the molecule,

$$\Delta p = (-mv_x) - (mv_x) = -2mv_x$$

After rebounding, this molecule travels towards opposite wall A_2 with velocity $-V_x$ collides to it and again rebounds with velocity V_x towards wall A_1.
Time between two successive collisions with the same wall,

$$\Delta t = \frac{\text{Distance travelled by molecule between two successive collisions}}{\text{Velocity of molecule}} = \frac{2L}{v_x}$$

Now force exerted by a single molecule on the shaded region of the wall is

$$F_{\text{Single molecule}} = \frac{\Delta p}{\Delta t} = \frac{2mv_x}{(2L / v_x)} = \frac{mv_x^2}{L}$$

Total force on the wall A_1 due to all the molecules,

$$F_x = \frac{m}{L}\sum v_x^2 = \frac{m}{L}(v_{x_1}^2 + v_{x_2}^2 + v_{x_3}^3 + \ldots) = \frac{mN}{L}(\overline{v_x^2})$$

$\overline{v_x^2}$ = Mean square of x component of the velocity

Now, pressure is defined as force per unit area, hence pressure on shaded wall,

$$P_x = \frac{F_x}{A} = \frac{mN}{AL}(\overline{v_x^2}) = \frac{mN}{V}(\overline{v_x^2})$$

For any molecule, the mean square velocity, $\overline{v^2} = \overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2}$;

By symmetry, we have,

$$\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2}$$

$$\Rightarrow \overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2} = \frac{\overline{v^2}}{3}$$

Total pressure inside the container,

$$P = \frac{1}{3}\frac{mN}{V}\overline{v^2} = \frac{1}{3}\frac{mN}{V}v_{rms}^2 \qquad \left(\text{where } v_{rms} = \sqrt{\overline{v^2}}\right)$$

RELATION BETWEEN PRESSURE AND TOTAL TRANSLATIONAL KINETIC ENERGY (E):

We can write $PV = \frac{1}{3}mNv_{rms}^2$

$$\Rightarrow PV = \frac{2}{3}\left(\frac{1}{2}mN\overline{v^2}\right)$$

$$\Rightarrow PV = \frac{2}{3}E \qquad \left[\text{Since } E = \frac{1}{2}(mN)\overline{v^2}\right]$$

$$\Rightarrow \frac{E}{V} = \frac{3}{2}P$$

So we can conclude that the total translational kinetic energy per unit volume (it is defined as energy density) of the gas is numerically equal to $\frac{3}{2}$ times the pressure exerted by the gas.

Also, we can write,

$$E = \frac{3}{2}PV$$

$$\Rightarrow E = \frac{3}{2}nRT = \frac{3}{2}NKT$$

$$\Rightarrow \frac{E}{N} = \frac{3}{2}KT$$

$\Rightarrow$ Average kinetic energy of a molecule is proportional to absolute temperature.

$$\frac{E}{N} = \frac{1}{2}m\overline{v^2} = \frac{3}{2}KT$$

$$\Rightarrow \overline{v^2} = \frac{3KT}{m}$$

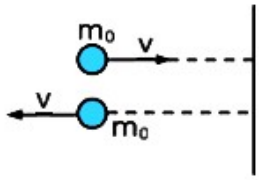
$$\Rightarrow v_{rms} = \sqrt{\frac{3KT}{m}} = \sqrt{\frac{3RT}{M}}$$

$$\Rightarrow v_{rms} \propto \sqrt{T} \text{ or } v_{rms} \propto \frac{1}{\sqrt{M_{mole}}}$$

Thus, at absolute zero the motion of all molecules of the gas stops.

Q. **In a period of 1.00 s, 5.00×10^{23} nitrogen molecules strike a wall with an area of 8.00 cm^2. Assume that the molecules move with a speed of 300 m/s and strike the wall head-on in elastic collisions. What is the pressure exerted on the wall?**
(The mass of one N_2 molecule is 4.65×10^{-26} kg)

Sol. To find the pressure exerted by the nitrogen molecules, we first calculate the average force exerted by the molecules.



$$\bar{F} = Nm_0 \frac{\Delta v}{\Delta t}$$

Here $N = 5.00 \times 10^{23}$, $m_0 = 4.65 \times 10^{-26}$ kg
$\Delta v = [300 - (-300)] = 2 \times 300$ m/s

$$= \frac{(5.00 \times 10^{23})[(4.65 \times 10^{-26} kg)2(300 m/s)]}{1.00 s} = 14.0 \text{ N}$$

Then pressure, $P = \frac{\bar{F}}{A} = \frac{14.0 N}{8.00 \times 10^{-4} m^2} = 17.5$ kPa

Q. **The rms value of a molecule of a given mass of a gas is 100 ms^{-1} at 27°C and at 1.00 atmospheric pressure. Find out the root mean square speeds of the molecules of the gas at 127°C and at 2.0 atmospheric pressure.**

Sol. We know that,

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\frac{(v_{rms})_1}{(v_{rms})_2} = \sqrt{\frac{T_1}{T_2}}$$

Given, $(v_{rms})_1 = 100$ m/s

$$T_1 = 27°C = (27 + 273)K = 300\ K$$

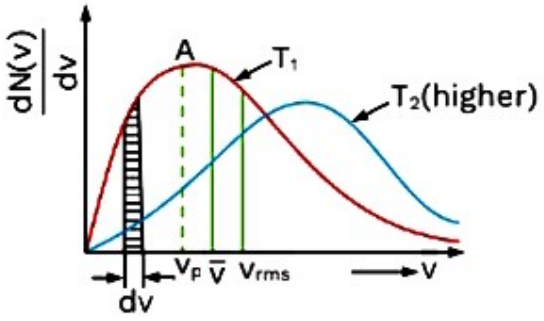
$$T_2 = 127°C = (127 + 273)K = 400\ K$$

$$\frac{100}{(v_{rms})_2} = \sqrt{\frac{300}{400}} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \quad (v_{rms})_2 = \frac{2 \times 100}{\sqrt{3}} = \frac{200}{\sqrt{3}}\ \text{m/s}$$

MAXWELL'S VELOCITY DISTRIBUTION LAW:

Distribution Curve: A plot of $\frac{dN(v)}{dv}$ (number of molecules per unit speed interval) against v is known as Maxwell's distribution curve.



$$\frac{dN}{dv} = 4\pi\left(\frac{m}{2\pi KT}\right)^{3/2} v^2 e^{-mv^2/2KT}$$

Now we can find the total Area under the curve by the integral $\int_0^\infty \frac{dN(v)}{dv}\,dv = \int_0^\infty dN(v) = N$.

The distribution curves show at two different temperatures. For any given temperature, the Area under the curve in that interval (shown with the shaded region) gives the number of molecules in a given speed interval dv. This number of molecules increases as the speed increases, up to a maximum value, and then decreases asymptotically towards zero value.

Thus, the maximum number of the molecules have speed lying within a small range centered about the speed corresponding to the peak (A) of the curve. This speed is corresponding the peak (A) of the curve. This speed is called the 'most probable speed' v_p or v_{mp}.
Maxwell's distribution curve is asymmetrical about the most probable speed v_p (its peak of distribution curve) because the lowest possible speed is zero, whereas there is no limit to the upper speed a molecule can attain.

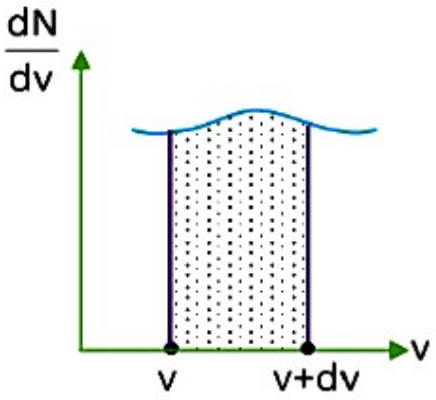
VELOCITY DISTRIBUTION AND CALCULATION OF AVERAGE SPEED & RMS SPEED:
Consider a container with total number of molecules. Now suppose dN is number of molecules moving with speed in range v to v + dv.

$$\frac{dN}{dv} = f(v)$$

$\Rightarrow$ $dN = f(v).dv$

$\Rightarrow$ $N_0 = \int dN = \int f(v).dv$

$\Rightarrow$ Total number of molecules = Area under given curve



Average Speed: It is defined as the arithmetic mean of the speeds of molecules in a gas at a given temperature.

$$v_{av} = \frac{v_1 + v_2 + v_3 + v_4 + \ldots}{N}$$

$$\text{Average speed} = \frac{\int dN.v}{\int dN} = \frac{\int f(v).vdv}{\int f(v).dv}$$

Average (or Mean) Speed: $\bar{v} = \sqrt{\frac{8kT}{\pi m}} = 1.59\ \sqrt{kT/m}$. (Derivation is not in the course)

Root mean square speed: It is defined as the square root mean of squares of the speed of different molecules.

i.e., $$v_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + v_4^2 + \ldots}{N}} = \sqrt{\overline{v^2}}$$

$$\text{R.M.S. Speed} = \left(\frac{\int dN v^2}{\int dN}\right)^{1/2} = \left(\frac{\int f(v).v^2 dv}{\int f(v) dv}\right)^{1/2}$$

RMS Speed: $$v_{rms} = \sqrt{< v^2 >} = \sqrt{\frac{3kT}{m}} = 1.73\sqrt{\frac{kT}{m}}$$

Most Probable Speed: The most probable speed (v_{mp}) is defined as the speed which is possessed by a maximum fraction of a total number of molecules of the gas. Mathematically, it is obtained by the condition.

$$\frac{df(v)}{dv} = 0$$

Most Probable Speed: $$v_{mp} = \sqrt{\frac{2P}{\rho}} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2kT}{m}}$$

Note that $v_{mp} < \overline{v} < v_{rms}$

Q. **Two molecules of a gas have speeds of 9×10^6 ms^{-1} and 1×10^6 ms^{-1}, respectively. What is the root mean square speed of these molecules?**

Sol. For n-molecules, we know that

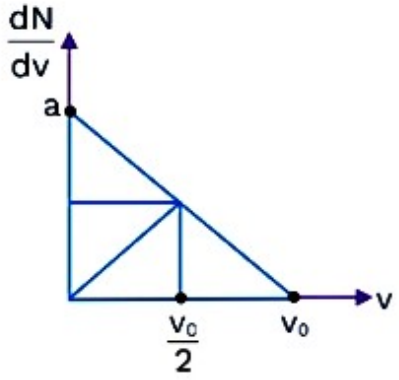
$$v_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + \ldots + v_n^2}{n}}$$

For two molecules, $$v_{rms} = \sqrt{\frac{v_1^2 + v_2^2}{2}}$$

Given, $v_1 = 9 \times 10^6$ m/s and $v_2 = 1 \times 10^6$ m/s

$$\therefore \quad v_{rms} = \sqrt{\frac{(9\times10^6)^2 + (1+10^6)^2}{2}} = \sqrt{41}\times10^6 \text{ m/s}$$

Q. **The graph shows a hypothetical speed distribution for a sample of N gas-particle-**



(i) Find the intercept (a) in terms of v_0 & N_0
(ii) Calculate R.M.S. Speed
(iii) What fraction of the total molecules are travelling with a speed greater than $\frac{v_0}{2}$.

Sol. (i) Area of graph = total molecules

$$\frac{1}{2}\times v_0 \times a = N$$

$$\Rightarrow \quad a = \frac{2N}{v_0}$$

(ii) $$v_{rms}^2 = \frac{\int v^2 dN}{\int dN}$$

We have, $$\frac{dN}{dv} = -\frac{a}{v_0}.v + a$$

$$\Rightarrow \quad dN = dv\left(\frac{-a}{v_0}v + a\right)$$

$$\Rightarrow \quad v_{rms}^2 = \frac{\int v^2 dN}{\int dN} = \frac{\int_0^{v_0} v^2 . \left(\frac{-av}{v_0} + a\right) . dv}{\int_0^{v_0} dv . \left(\frac{-av}{v_0} + a\right)} \Rightarrow v_{rms}^2 = \frac{\left[-\frac{v_0^3}{2} + \frac{v_0^3}{3}\right]}{\left[\frac{-v_0}{2} + v_0\right]} = \frac{v_0^2}{6}$$

$$\Rightarrow \quad v_{rms} = \frac{v_0}{\sqrt{6}}$$

(iii) Number of molecules having speed greater than $\frac{v_0}{2} = \int_{\frac{v_0}{2}}^{v_0} f(v)dv$

$= \frac{3}{4}$ Area of given triangle

$= \frac{3}{4}\left(\frac{1}{2} a v_0\right)$

$= \frac{3}{8} a v_0$

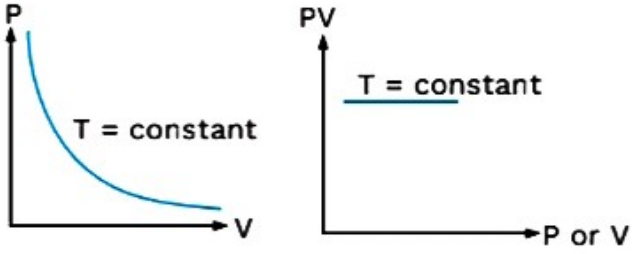
$= \frac{3}{4} N_0$

Gas Laws

DERIVATION OF GAS LAWS FROM KINETIC THEORY OF GASES:

BOYLE'S LAW:

According to Boyle's law, at a given temperature, the pressure of a given mass of a gas is inversely proportional to its volume. This is known as Boyle's law.



$$PV = \frac{1}{3}mNv_{rms}^2$$

As for a given gas $v_{rms}^2 \propto T$, the value v_{rms}^2 is constant at a given temperature.

PV = Constant

$$P \propto \frac{1}{V}$$ ...**Boyle's Law.**

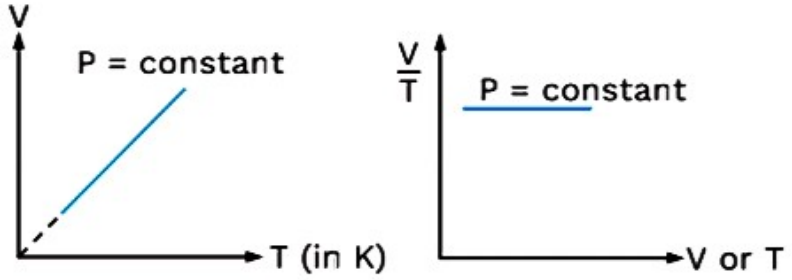
Also, $PV = P\left(\frac{m}{\rho}\right)$ = constant (m = mass of the gas and ρ is the density of gas)

$\Rightarrow \frac{P}{\rho}$ = constant or $\frac{P_1}{\rho_1} = \frac{P_2}{\rho_2}$

Thus, P-V graph is a rectangular hyperbola or PV versus P or V graph is a straight line parallel to P or V axis.

CHARLES' LAW:

According to Charles law, if the pressure of the gas is constant, then the volume of a given mass of a gas is proportional to its absolute temperature. This is known as Charles' Law.



From the kinetic theory of gases, if P is constant, $V \propto v_{rms}^2$.

As $v_{rms}^2 \propto T$, we get $V \propto T$ which is Charles' law.

$$\Rightarrow \frac{V}{T} = \text{constant}$$

$$\Rightarrow \frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{V}{T} = \frac{m}{\rho T} = \text{constant} \qquad \left(\text{As volume } V = \frac{m}{\rho}\right)$$

or $\rho T = \text{constant}$

$\Rightarrow \rho_1 T_1 = \rho_2 T_2$

Thus, V–T graph is a straight line passing through origin or V/T versus V or T graph is a straight line parallel to V or T axis.

AVOGADRO'S LAW:

At the same temperature and pressure, equal volumes of all gases contain an equal number of molecules. This is known as Avogadro's law.

From kinetic theory of gases, we have

$$PV = \frac{1}{3}N_1 m_1 v_1^2 \text{ and } PV = \frac{1}{3}N_2 m_2 v_2^2 \qquad ...(1)$$

Since, the average kinetic energy of the molecules is same for the two gases, at a given temperature,

$$\frac{1}{2}m_1 v_1^2 = \frac{1}{2}m_2 v_2^2 \qquad ...(2)$$

From equations (1) and (2), $N_1 = N_2$, which proves Avogadro's law.

DALTON'S LAW OF PARTIAL PRESSURE:

Dalton's law of partial pressure says that the pressure exerted by a mixture of several gases equals the sum of the pressure exerted by each gas occupying the same volume as that of the mixture.

In kinetic theory, we assume that the pressure exerted by a gas on the walls of a container is due to the collisions of the molecules with the walls. The total force on the wall is the sum of the forces exerted by the individual molecules. Suppose there are N1 molecules of gas 1, N2 molecules of gas 2, etc., in the mixture.

Thus, the force on a wall of surface area A, F = force by N_1 molecules of gas 1 + force by N_2 molecules of gas 2 +... = $F_1 + F_2$ +...

Thus, the pressure,

$$P = \frac{F_1}{A} + \frac{F_2}{A} + \ldots$$

$$P = P_1 + P_2 + P_3 + \ldots$$

$$PV = (n_1 + n_2 + \ldots)\, RT$$

Q. **A vessel of volume 1660 cm³ contains 0.1 moles of oxygen and 0.2 moles of nitrogen. Find its pressure, if the temperature of the mixture is 300 K.**

Sol. We have $PV = nRT \Rightarrow P = \frac{nRT}{V}$

The partial pressure due to oxygen,

$$P_1 = \frac{(0.1\text{mol})(8.3\,\text{J/mol} - \text{K})(300\text{K})}{(1660 \times 10^{-6}\,\text{m}^3)} = 1.5 \times 10^5\ \text{Pa}$$

Similarly, the partial pressure due to nitrogen, $P_2 = 3.0 \times 10^5$ Pa.
The total pressure in the vessel,

$$P = P_1 + P_2 = (1.5 + 3.0) \times 10^5\ \text{Pa} = 4.5 \times 10^5\ \text{Pa}$$

EQUATION OF STATE OR IDEAL GAS EQUATION:
Experimenters have found that at low enough densities, all real gases act as an ideal gas and tend to obey the following relation.

$$PV = nRT \quad \text{...}\textbf{equation of state}$$

Here, **R** is a constant called the gas constant that has the same value for all gases-namely, R = 8.31 J/mol-K.
In alternative form: PV = NKT
Where K is **Boltzmann constant** and N is a number of molecules.

$$K = \frac{R}{N_A} = \frac{8.31\,\text{J/mol} - \text{K}}{6.02 \times 10^{23}\,\text{mol}^{-1}} = 1.38 \times 10^{-23}\ \text{J/K}$$

Important Points:
(i) One mole of an ideal gas at STP has a volume of 22.4 litres (= 22.4×10^{-3} m³).
(ii) At higher temperature and low pressure or at higher temperature and low density, a real gas behaves as an ideal gas.

Q. **(a) How many atoms of helium gas fill a spherical balloon of diameter 30.0cm at 20.0°C and 1 atm?**
(b) Find out the average kinetic energy of the helium atoms?
(c) What is the rms speed of the helium atoms?

Sol. (a) The volume of helium gas in the balloon,

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(0.150\,m)^3 = 1.41 \times 10^{-2}\ m^3$$

Number of moles can be obtained from ideal gas equation,

$$n = \frac{PV}{RT}$$

$$n = \frac{(1.013 \times 10^5\ N/m^2)(1.41 \times 10^{-2}\ m^3)}{(8.314\,N.m/mol - K)(293\,K)} = 0.588\ mol$$

The number of molecules of helium gas,

$$N = nN_A = (0.588\ mol)\ (6.02 \times 10^{23}\ molecules/mol)$$

$$\Rightarrow \quad N = 3.54 \times 10^{23}$$

(b) The average kinetic energy,

$$\bar{K} = \frac{1}{2}m_0 v_{rms}^2 = \frac{3}{2}k_B T$$

$$\Rightarrow \quad \bar{K} = \frac{3}{2}(1.38 \times 10^{-23}\ J/K)(293K) = 6.07 \times 10^{-21}\ J$$

(c) The mass of an atom of helium gas,

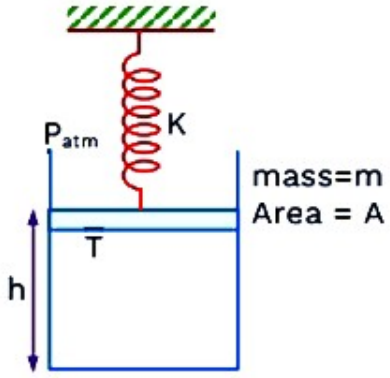
$$m_0 = \frac{M}{N_A} = \frac{4.0026\,g/mol}{6.02 \times 10^{23}\ molecules/mol}$$

$$= 6.65 \times 10^{-24}\ g = 6.65 \times 10^{-27}\ kg$$

So, the root-mean square speed, $v_{rms} = \sqrt{\frac{2K}{m_0}}$

$$\Rightarrow \quad v_{rms} = \sqrt{\frac{2 \times 6.07 \times 10^{-21}\ J}{6.65 \times 10^{-27}\ kg}} = 1.35\ km/s.$$

Q. **A non-conducting piston of mass m and Area of cross-section A is placed on a non-conducting cylinder, as shown in the figure. The height, temperature, and spring constant of the piston are given by h, T, K, respectively. It is given that initially, spring is relaxed, and the piston is at rest. Then calculate the following parameters**



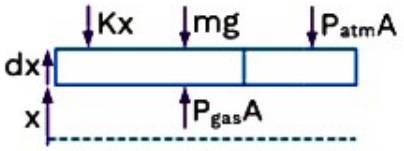
(i) Number of moles.
(ii) when the gas is heated, slowly find the Work done by gas to displace the piston by distance d.

Sol. (i) $PV = nRT$

$$\Rightarrow \left(P_{atm} + \frac{mg}{A}\right)Ah = nRT$$

$$\Rightarrow n = \frac{\left(P_{atm} + \frac{mg}{A}\right)Ah}{RT}$$

(ii) Applying Newton's law on the piston $mg + P_{atm}A + Kx = P_{gas}A$



$$W_{gas} = \int_0^d P_{gas}Adx \quad \Rightarrow \quad W_{gas} = \int_0^d (mg + P_{atm}A + Kx)\,dx$$

$$\Rightarrow W_{gas} = mgd + P_{atm}Ad + \frac{1}{2}Kd^2$$

Internal Energy

DEGREE OF FREEDOM:

Degree of freedom can be defined as the number of independent expressions required to completely define the motion of a body is called its degree of freedom. Note that motion along all the axes must be independent of each other.

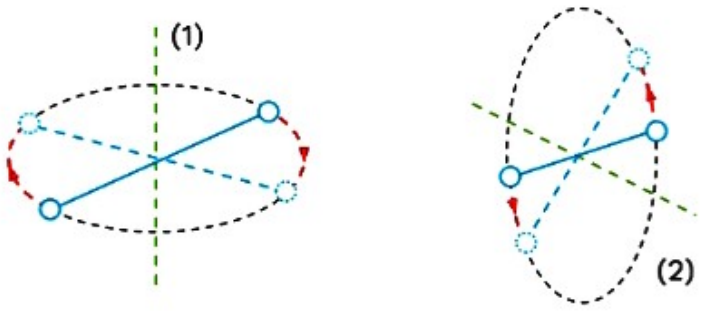
TRANSLATIONAL DEGREES OF FREEDOM:

The translational motion can be defined as the motion of a body from one point to another. Thus, a molecule that is free to move in space has three degrees of freedom because a point object can move in x, y, and z directions independently. If In the given tiny object has constrained to move in a plane, then this object has two degrees of freedom, and if it is constrained to move in a line, then its degree of freedom is one.
For Mono Atomic gases (all inert gases, e.g., He, Ar, etc.), f = 3(translational)

ROTATIONAL DEGREES OF FREEDOM:

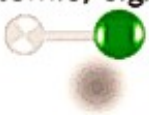
In addition to a translational degree of freedom, the diatomic molecule can also rotate about its center of mass. In the given Figure, it shows that the two independent axes of rotation 1 and 2, perpendicular to the axis of joining the two oxygen atoms. Oxygen molecules can rotate about this axis. So, according to the definition of degree of freedom, the molecule thus has two rotational degrees of freedom, each of which contributes a term to the total energy consisting of translational energy and rotational energy.

$$K.E = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 + \frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2$$



The two independent axes of rotation of a diatomic molecule where ω_1 and ω_2 are the angular speeds about the axes 1 and 2 and I_1, I_2 are the corresponding moments of inertia. Note that each rotational degree of freedom contributes a term to the energy that contains square of a rotational variable of motion.

For Diatomic gases like H_2, N_2, O_2 etc., f = 5 (3 translational + 2 rotational)

Type of molecule	N	Degrees of freedom	Mode
Monatomic, e.g. Ne 	1	3	3 translational 0 rotational 0 vibrational
Diatomic, e.g. HCl 	2	6	3 translational 2 rotational 1 vibrational
Triatomic linear, e.g. CO_2 	3	9	3 translational 2 rotational 4 vibrational
Triatomic non-linear, e.g. H_2O 	3	9	3 translational 3 rotational 3 vibrational

MAXWELL'S LAW OF EQUIPARTITION OF ENERGY:

Energy associated with each degree of freedom = $\frac{1}{2}kT$. The total kinetic energy of that molecule U = $\frac{1}{2}fkT$; where f is the degree of freedom of a molecule. That means, in equilibrium conditions, the total energy of a molecule or atom is equally distributed in all possible energy modes, with each mode having average energy equal to 1/2KT.

Degree of freedom for Mono Atomic gas: (all inert gases Ex. He, Ar etc.) f = 3(translational)

$$K.E = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 = 3/2\ KT$$

Degree of freedom for Diatomic gas: (gases like H_2, N_2, O_2 etc.) f = 5 (3 translational + 2 rotational)

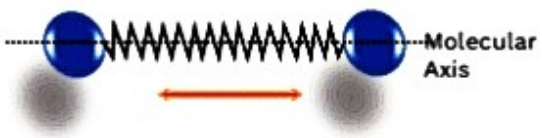
$$K.E = \frac{1}{2}m\upsilon_x^2 + \frac{1}{2}m\upsilon_y^2 + \frac{1}{2}m\upsilon_z^2 + \frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2 = 5/2KT$$

Kinetic Energy K.E = $\frac{f}{2}$NKT

...where f is the total degree of freedom and N is the number of molecules.

VIBRATIONAL DEGREE OF FREEDOM AND ENERGY ASSOCIATED WITH IT:

We have assumed above that the O2 molecule is a 'rigid rotator,' i.e., the molecule does not vibrate. This assumption, though found to be true (at moderate temperatures) for O2, is not always valid. Molecules like CO, even at moderate temperatures, have a mode of vibration, i.e., their atoms oscillate along the interatomic axis like a one-dimensional oscillator and contribute a vibrational energy term E0 to the total energy:



$$E_o = \frac{1}{2}m\left(\frac{dy}{dt}\right)^2 + \frac{1}{2}ky^2$$

Where k is the force constant of the oscillator and y the vibrational co-ordinate. Once again, the vibrational energy terms in the equation contain squared terms of vibrational variables of motion y and dy/dt.

While each translational and rotational degree of freedom has contributed only one 'squared term' in the equation and 1/2KT energy, one vibrational mode contributes two 'squared terms' and hence 2(1/2KT) = KT: kinetic and potential energies

Vibrational degree of freedom (f_v) = 3N – Sum of translational and rotational degree of freedom

$$f_v = 3N - (f_t + f_r)$$

INTERNAL ENERGY(U):

The internal energy of a system is the sum of potential energies and kinetic of the molecules of the system.

Internal energy (U) of the system is the function of its volume (V) and its absolute temperature (T).

Since intermolecular force is zero in the case of an ideal gas, this its potential energy is also zero. In this case, the internal energy is only due to kinetic energy, which depends on the absolute temperature of the gas, i.e., U = f(T).

For an ideal gas internal energy U can be written as = $\frac{f}{2}nRT$

$$U = (f_t + f_r)\ \frac{1}{2}nRT\ + [3N - (f_t + f_r)]\ nRT$$

EQUIVALENT DEGREES OF FREEDOM FOR A GASEOUS MIXTURE:

If a number of gases with degrees of freedom f_1, f_2, f_3,...f_N are mixed with n_1, n_2, n_3,..., n_N moles at the same temperature T, then their total internal energy before mixing can be given as

$$U_i = \frac{f_1}{2}n_1RT + \frac{f_2}{2}n_2RT + ... + \frac{f_N}{2}n_NRT \qquad ...(i)$$

And after mixing, the internal energy of this mixture can be given as

$$U_f = \frac{f_{eq}}{2}(n_1 + n_2 + ... + n_N)RT \qquad ...(ii)$$

As no energy loss is taking place during mixing of gases.

$$U_i = U_f$$

$$\frac{f_1}{2}n_1RT + \frac{f_2}{2}n_2RT + ... + \frac{f_N}{2}n_NRT\ = \frac{f_{eq}}{2}(n_1 + n_2 + ... + n_N)RT$$

$$\Rightarrow f_{eq} = \frac{f_1n_1 + f_2n_2 + ...f_Nn_N}{n_1 + n_2 + ...n_N} \qquad ...(iii)$$

MIXING OF GASES AT CONSTANT VOLUME:

When some gases at different temperatures are mixed at constant volume in a thermally insulated vessel, the total internal energy of all the gases remains constant. Before mixing internal energy of the gas is

$$U_i = \frac{f_1}{2}n_1RT_1 + \frac{f_2}{2}n_2RT_2 + ... + \frac{f_N}{2}n_NRT_N$$

If after mixing, temperature of mixture becomes T_f, then the total internal energy of gas after mixing of gases is

$$U_f = \frac{f_1}{2}n_1RT_f + \frac{f_2}{2}n_2RT_f + ... + \frac{f_N}{2}n_NRT_f$$

As total internal energy of the gaseous mixture must remain constant, thus

$$U_i = U_f$$

$$\Rightarrow T_f = \frac{f_1n_1T_1 + f_2n_2T_2 + ... + f_Nn_NT_N}{f_1n_1 + f_2n_2 + ... + f_Nn_N}$$

Q. **A light container contains n mole diatomic gas. Mass of gas is M. The container is moving with velocity V.**
(i) Find the kinetic energy of gas with respect to the center of mass of the system.
(ii) What is KE of gas w.r.t. ground?

Sol. (i) K.E. $= \frac{5}{2}nRT$ $\boxed{\begin{matrix}\text{mass of gas} = M \\ \text{temperature } T\end{matrix}} \longrightarrow V$

(ii) Kinetic energy of gas w.r.t. ground = Kinetic energy of gas w.r.t. centre of mass + Kinetic energy of centre of mass w.r.t. ground.

$$\text{K.E.} = \frac{1}{2}Mv^2 + \frac{5}{2}nRT$$

Q. **Consider a closed rigid container having 2 moles of diatomic gas at temperature 'T'. If temperature is doubled, 50% of molecules get dissociated in atomic form. Calculate increment in K.E. of molecules and the ratio of final & initial pressure.**

Sol. $2A_2 \longrightarrow A_2 + 2A$

2 moles of A_2 1 mole of A_2 + 2 mole of A

Initial Internal Energy $(U_i) = \frac{5}{2}(2)RT = 5RT$

Final Internal Energy $(U_f) = \frac{5}{2}(1)R2T + \frac{3}{2}(2)R(2T)$

$= 11\ RT$

$\Rightarrow$ Change in Kinetic Energy $= U_f - U_i = 6RT$

However, change in internal energy would be greater than 6RT as some energy is invested in breaking bond.

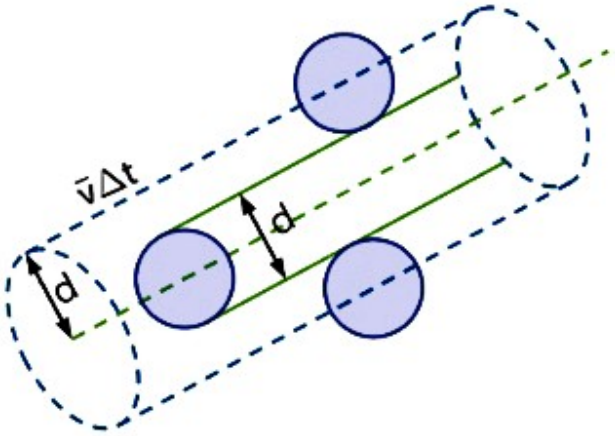
Now, $P_1V = 2RT$

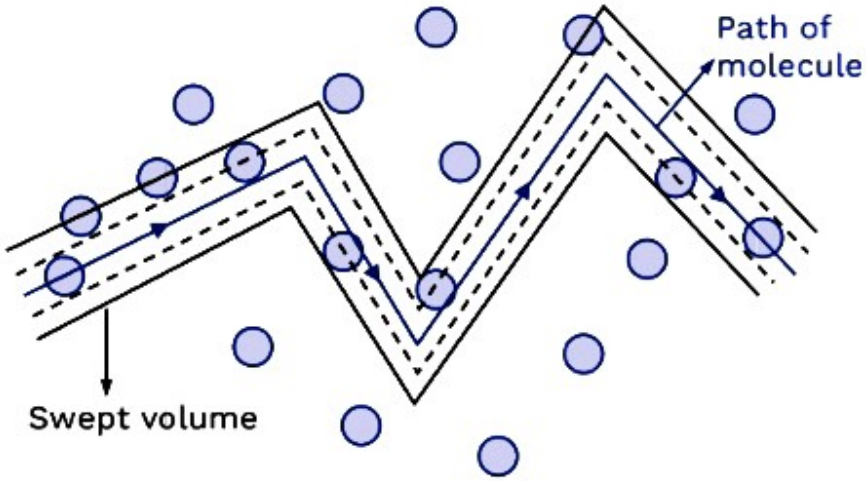
$$P_2V = 3R(2T)$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{2}{6} = \frac{1}{3}$$

MEAN FREE PATH:

Molecules in a gas have rather large speeds, yet a gas leaking from a cylinder in a kitchen takes considerable time to diffuse to the other corners of the room. This happens because molecules in a gas have a finite though small size, so they are bound to undergo collisions. As a result, they cannot move straight unhindered; their paths keep getting continuously deflected.





Suppose the molecules of a gas are spheres of diameter d. Focus on a single molecule with the average speed <v>. It will suffer a collision with any molecule that comes within a distance d between the centres. In time Δt, it sweeps a volume

$\pi d^2 \langle v \rangle \Delta t$ wherein any other molecule will collide with it.

If n is the number of molecules per unit volume, the molecule suffers $n\pi d^2 \langle v \rangle \Delta t$ collisions in time Δt. Thus, the rate of collisions is $n\pi d^2 \langle v \rangle$ or the time between two successive collisions is on the average, called the **mean free time** is:

$$\tau = 1/(n\pi \langle v \rangle d^2)$$

The average distance between two successive collisions, called the **mean free path** is:

$$\ell = \langle v \rangle \tau = 1/(n\pi d^2)$$

$$\ell = \langle v \rangle \tau = \frac{1}{\sqrt{2}\pi\left(\frac{N}{V}\right)d^2}$$

(where N be number of molecules in volume V of container)

Here, we assumed that the other molecules were at rest. But in reality, all molecules are moving, and the collision rate is determined by the average relative velocity of the molecules. Thus, we need to replace $\langle v \rangle$ by $\langle v_r \rangle$.

$$\ell = \frac{1}{\sqrt{2}nd^2}$$

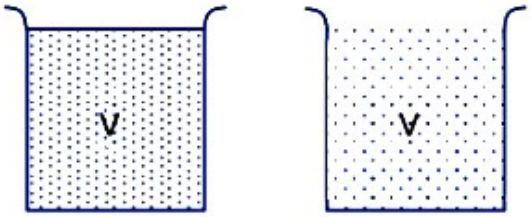
It is also expected that the mean free path given by the equation depends inversely on the number density and the size of the molecules. In a highly evacuated tube, n is rather small,

and the mean free path can be as large as the length of the tube.

Q. **Consider two containers of the same volume, pressure, and temperature first is closed one and 2nd is open. If temperature is doubled in both the containers, then by what factor following quantities will change in each container?**
(i) K.E. of molecules in container
(ii) R.M.S. speed
(iii) Mean free path
(iv) Relaxation time (average time between two collisions)

Sol. Let initial number of moles in containers be n.

So, $n = \frac{PV}{RT}$



Container 1 Container 2

Finally, number of moles in closed container remains same and number of moles in open container

$$n_2 = \frac{PV}{R(2T)} = \frac{n}{2}$$

(i) Kinetic energy in closed containser $= \frac{f}{2}nRT$

Kinetic energy in open container $= \frac{f}{2}(2n)R\left(\frac{T}{2}\right)$

$$= \frac{f}{2}nRT$$

So, kinetic energy will remain same in both the container.

(ii) RMS speed depends only on temperature, so if temperature doubles,

$= v_{rms} \propto \sqrt{T}$

$$\frac{v_{rms_1}}{v_{rms_2}} = \sqrt{\frac{2T}{T}} = \sqrt{2}$$

So, v_{rms} becomes $\sqrt{2}$ times.

(iii) ℓ (mean free path) $= \frac{1}{\sqrt{2}\pi\left(\frac{N}{V}\right)d^2}$

$$\frac{\ell_2}{\ell_1} = \frac{N_1}{N_2} = \frac{n}{n/2} = 2$$

So, mean free path (ℓ) becomes 2 times.

(iv) So, $\tau \propto \frac{\ell}{v}$

$$\Rightarrow \quad \frac{\tau_2}{\tau_1} = \frac{\ell_2}{\ell_1} \times \frac{v_1}{v_2} = 2 \times \frac{1}{\sqrt{2}}$$

$$\Rightarrow \quad \frac{\tau_2}{\tau_1} = \sqrt{2}$$

So, relaxation time becomes $\sqrt{2}$ times.